

**AUTOMATED TRASH BIN**

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Republic of the Philippines

**CAPIZ STATE UNIVERSITY**  
**DAYAO SATELLITE COLLEGE**

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In partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science**, this thesis entitled “**AUTOMATED TRASH BIN**”, prepared and submitted by **Paul Anthony Cerilo, Arthur Ballonado, Vishnu Crist Dela Cruz, Janna Marie Adrimesen** and **Nicole Marie Clavel** has been evaluated for final oral examination.

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## ABSTRACT

**Paul Anthony Cerilo, Arthur Ballonado, Vishnu Crist Dela Cruz, Janna Marie Adrimesen and Nicole Marie Clavel**, Capiz State University, Dayao Campus, Dayao Roxas City, June 2025. **Automatic Trash Bin.**

Thesis Adviser: **RIO L. VERGABERA, ME**

Waste management and environmental sustainability is very necessary nowadays, especially that we are prone to virus attacks and alike.

This study aimed to develop an Automated Trash Bin for Capiz State University Dayao Satellite College specifically to the different offices of the college in order to improve the effectiveness of waste segregation and promote environmental sustainability in the community.

The primary goal of this project is to promote efficient waste management by reducing manual sorting, encouraging proper disposal habits, and integrating technological advancements. By automating the process, the system aims to enhance environmental sustainability while making waste disposal more convenient for users.

The study employs a developmental and experimental research design, focusing on the design, implementation, and evaluation of an automated trash bin system. The research will involve prototype development and performance testing to assess efficiency in waste segregation.

The implications of an automated trash bin extend across environmental, social, and technological dimensions.

Results indicate that automated waste segregation systems can significantly reduce contamination rates, optimize recycling processes, and contribute to more sustainable waste management practices. Furthermore, the study evaluates the environmental and economic benefits of adopting such systems at scale, offering valuable insights for future development and implementation in both urban and industrial settings. The system was

rated 4.10, with a verbal interpretation of "Acceptable," indicating that the system is effective and meets ISO standards for software quality.

Keywords: *Automated, Trash Bin, Ultra sonic*

